

Efrat Dener, Ph.D.

Plant ecology · seed dispersal & dispersal evolution · intraspecific trait variation · plant–pollinator interactions · statistical & computational ecology

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269 citations · h-index 6 · i10-index 5 (Google Scholar, 2026). Journal impact factors: most recent JCR (2024).

RESEARCH FOCUS

I study how ecological and evolutionary forces shape variation in plant traits — especially seed dispersal — across organizational and spatial scales. A recurring role in my work is that of the **integrative quantitative analyst** on large, multi-year projects: I synthesize data generated by multiple students across distinct sub-studies into unified hierarchical, cross-scale analyses, supported by custom analytics dashboards. This integrative approach spans the multi-chapter dispersal-trait project with PhD student M. García Serrano, the Asiatic wild-ass ecology project, and my own doctoral research.

EDUCATION

Ph.D., Desert Studies / Ecology 2016–2021

Albert Katz School for Desert Studies, Mitrani Dept. of Desert Ecology, Ben-Gurion University (BGU)

Thesis: The interactive effects of genetic relatedness and selection pressures on seed dispersal distances in an annual plant. Supervisors: Prof. Itamar Giladi & Prof. Ofer Ovadia.

M.Sc., Desert Ecology — Cum Laude 2014–2016

Albert Katz School for Desert Studies, BGU

Thesis: The effect of pollen source on dispersal-related traits in a heterocarpic annual plant.

Supervisors: Dr. Itamar Giladi & Dr. Hagai Shemesh.

B.Sc., Environmental Sciences 2010–2014

Tel-Hai Academic College

Research project: The effect of temporally variable nutrient regimes on habitat selection in an annual plant. Supervisor: Dr. Hagai Shemesh.

ACADEMIC APPOINTMENTS

Postdoctoral Researcher — Ben-Gurion University 2022–present

Spatial Plant Ecology Lab (Prof. Itamar Giladi) — intraspecific trait variation, dispersal ecology, hierarchical & variance-partitioning modeling. Plant Communities & Complex Systems Lab (Prof. Merav Seifan) — plant–pollinator interactions, drought effects on plant fitness.

Postdoctoral Researcher — Tel-Aviv University 2021

Lab of Prof. Itay Mayrose — effects of genomic duplication (polyploidy) on plant–pollinator interactions and ecological-network structure (phylogenetic comparative methods, network theory). Collaboration with Prof. Tia-Lynn Ashman (Univ. of Pittsburgh).

PUBLICATIONS

Author legend: **Dener, E.** = author; † = equal/co-first contribution. Reverse chronological.

Peer-reviewed journal articles

12. García, M.J., **Dener, E.** & Giladi, I. (2026). Multi-level variation in dispersal traits along a precipitation gradient. *Global Ecology and Conservation*, e04246. IF 3.7

11. Paul, S., Avidan, Y., Shahar, N., **Dener, E.**, Bar-David, S. & Seifan, M. (2026). Dietary insights into ecological integration and impact on vegetation of a reintroduced large herbivore in a hyper-arid ecosystem. *Global Ecology and Conservation*, e04215. IF 3.7
10. Salman, I.N.A., **Dener, E.**, Tzin, V. & Seifan, M. (2025). Plant fitness is shaped by cascading effects of aridity and drought on floral traits and pollination services. *Agriculture, Ecosystems & Environment* 393, 109855. IF 6.4 · 1 cit.
9. Krintza, N., **Dener, E.** & Seifan, M. (2024). Stress induces trait variability across multiple spatial scales in the arid annual plant *Anastatica hierochuntica*. *Plants* 13(2), 256. IF 4.1 · 7 cit.
8. Shemesh, H., **Dener, E.** & Sadeh, A. (2022). Bedrock may dictate the distribution of the fire salamander in the southern border of its global range. *Israel Journal of Ecology and Evolution* 69(1–2), 5–9. IF 0.8 · 3 cit.
7. Pozner, E., Bar-On, P., Livne-Luzon, S., Moran, U., Tsamir-Rimon, M., **Dener, E.**, et al. (2022). A hidden mechanism of forest loss under climate change: the role of drought in eliminating forest regeneration at the edge of its distribution. *Forest Ecology and Management* 506, 119966. IF 3.7 · 49 cit.
6. Greenbaum, G., **Dener, E.** & Giladi, I. (2022). Limits to the evolution of dispersal kernels under rapid fragmentation. *Journal of the Royal Society Interface* 19(188), 20210696. IF 3.5 · 1 cit.
5. Cna'ani, A.†, **Dener, E.**†, Ben-Zeev, E., Günther, J., Köllner, T.G., Tzin, V. & Seifan, M. (2021). Phylogeny and abiotic conditions shape the diel floral emission patterns of desert Brassicaceae species. *Plant, Cell & Environment* 44(8), 2656–2671. IF 6.3 · 10 cit. († joint first authors)
4. **Dener, E.**, Shemesh, H. & Giladi, I. (2021). The effect of pollen source on seed traits and dispersability in the heterocarpic annual *Crepis sancta*. *Journal of Plant Ecology* 14(3), 425–437. IF 3.9 · 6 cit.
3. **Dener, E.**, Ovadia, O., Shemesh, H., Altman, A., Chen, S.-C. & Giladi, I. (2021). Direct and indirect effects of fragmentation on seed dispersal traits in a fragmented agricultural landscape. *Agriculture, Ecosystems & Environment* 309, 107273. IF 6.4 · 30 cit.
2. Chen, S.-C., **Dener, E.**, Altman, A., Chen, F. & Giladi, I. (2020). Effect of habitat fragmentation on seed dispersal ability of a wind-dispersed annual in an agroecosystem. *Agriculture, Ecosystems & Environment* 304, 107138. IF 6.4 · 16 cit.
1. **Dener, E.**, Kacelnik, A. & Shemesh, H. (2016). Pea plants show risk sensitivity. *Current Biology* 26(13), 1763–1767. IF 7.5 · 146 cit. *Widely covered in international science media (Science/AAAS, The Scientist, ScienceDaily, Phys.org, Univ. of Oxford).*

Preprints / manuscripts under review

- Livne-Luzon, S., Yaakobi, A., Yalin, D., Sade, D., **Dener, E.**, et al. (2026). Tree–microbe–soil interactions affecting soil organic carbon fractions in Mediterranean forest soils. *EGUsphere* preprint (submitted to *Journal of Ecology*).
- Ibdah, M., Kedoshim, R., Nawade, B., Cohen, Y., **Dener, E.**, Revuru, B. & Shaltiel-Harpaz, L. Pollination protects against herbivory: unintended consequences of pesticides and volatile signal manipulation. Submitted to *Journal of Applied Ecology* (2026). *Contribution: SEM, bootstrap estimation of indirect effects.*
- Ostreiher, R., **Dener, E.** & Heifetz, A. Time perception guides foraging decisions in free-living Arabian babblers. *Animal Behavior and Cognition* (in revision, 2026). *Contribution: statistical methods, results, figures.*

Manuscripts in preparation

- **Dener, E.**, García, M.J. & Giladi, I. Environmental effects on variation in dispersal traits across organizational levels. (*First author.*)
- **Dener, E.** & Giladi, I. Dispersal traits in urban vs. non-urban environments across Israeli cities.

- Avidan, Y., Bar-David, S., Altman, A., **Dener, E.** & Seifan, M. No long-term effects of reintroduced large herbivores on vegetation composition in an arid region. (*joint species distribution modeling + variance partitioning.*)

RESEARCH IN THE MEDIA

Dener et al. (2016), "Pea plants show risk sensitivity," Current Biology — covered by 20+ international outlets following Oxford and Cell Press releases, including **Science/AAAS**, *Newsweek*, *Christian Science Monitor*, *The Daily Beast*, *The Scientist*, IFLScience, Modern Farmer, UPI, Phys.org, ScienceDaily; German outlets *wissenschaft.de* and *scinexx.de*; and the Hebrew **Davidson Institute** (Weizmann). Discussed in a *Current Biology* Dispatch commentary, "Decision-Making: Are Plants More Rational than Animals?" (2016).

RESEARCH FUNDING & GRANT CONTRIBUTIONS

2020	ISF Grant 2684/20 (contributing author) — "Levels of intraspecific variation in dispersal traits and their potential consequences" (PI: Prof. I. Giladi). ~848,000 NIS / 4 yrs. Created proposal figures and contributed written feedback across drafts. Funds the co-supervised PhD project.
2024	ISF Grant 2886/25 (contributing collaborator) — "Dust devils as a mechanism of secondary seed dispersal in drylands" (PI: Prof. I. Giladi). Proposal design, graphics, methods information.
2014	The Sol Leshin Grant — 87,000 NIS. Co-initiated, developed and co-wrote (with Dr. I. Giladi & Dr. H. Shemesh).

AWARDS & HONORS

2021	Publication Prize ×2 (for <i>Dener et al.</i> , <i>J. Plant Ecology</i> ; and <i>Cna'ani & Dener et al.</i> , <i>Plant, Cell & Environment</i>) — Albert Katz Int'l School for Desert Studies, BGU
2021	Postdoctoral Matching Scholarship — Tel-Aviv University
2018	Second-Best Student Presentation Award — 31st GfÖ Plant Population Biology Conference
2016–2020	The Negev (Faran) Scholarship for Outstanding Doctoral Students — BGU
2016	Best Student Presentation Award — 29th GfÖ Plant Population Biology Conference
2016	The Merav Ziv Award for Outstanding Achievement in Ecology (M.Sc.) — BGU
2016	"Recognition of Excellence" Prize; Dean's Award (Certificate of Merit) — Albert Katz Int'l School, BGU
2015–2016	Full M.Sc. Scholarship — Albert Katz School for Desert Studies, BGU

CONFERENCE PRESENTATIONS

2025	Multi-level variation in dispersal traits along a precipitation gradient — PopBio Annual Conference (Poster)
2020	Effect of habitat fragmentation on seed dispersal ability of a wind-dispersed annual — 4th NSO / Oikos Conference, Reykjavik (Poster)
2019	Evolution of dispersal kernels under anthropogenic fragmentation — 4th Plant Ecology in Israel, Tel-Hai (Oral)
2018	Dispersal in fragmented habitats — 3rd Plant Ecology in Israel, Tel-Aviv Univ. (Oral); & 31st GfÖ Conf., Innsbruck (Oral)

2016	Long-distance pollination results in short-distance seed dispersal — 29th GfÖ Conf., Czech Rep. (Oral); 17th Sede Boqer Symposium (Oral); 1st Plant Ecology in Israel (Oral)
2015	The effect of pollen source on dispersal-related traits — Israeli Society of Ecology & Environmental Science, Hebrew Univ. (Oral)
2014	Beyond average: temporally variable nutrient regimes and habitat selection — ISEES, Bar-Ilan Univ. (Oral)

TEACHING EXPERIENCE

Lecturer (Instructor of Record) — Tel-Hai Academic College

2020–2022	Statistics for Biologists / Biostatistics (Biotechnology & Water Science)
2020–2021	Ecological Tracking as a Research Tool (Environmental Sciences)
2021–2022	Research Methods B (Environmental Sciences)
2022	First Seminar (Environmental Sciences) — co-taught

Teaching Assistant / Guest Lecturer — Ben-Gurion University

2022–pres.	Data Visualization module, Biostatistics course (guest sessions)
2019–2021	The World of Plants (Life Sciences Dept.)
2017–2019	Biophysical Field Methods in the Namib Desert — international course, Gobabeb, Namibia (in English)
2020	Biostatistics & Experimental Design, Albert Katz Int'l School (in English)

MENTORING & STUDENT SUPERVISION

2022–pres.	Manuel Jesús García Serrano — Ph.D. (BGU) — co-supervisor with Prof. Giladi
2023–2026	Ibrahim Salman — Ph.D. (BGU) — statistical & manuscript mentor; co-author
2024–2026	Hemanth Akallasandra Yoganan — M.Sc. (BGU) — statistical analysis & supervision
2026	Sanjana Sarode — M.Sc. — thesis mentoring
2026	Nuribat Cohen & Efrat Levi — seminar students (BGU, Seifan group) — co-mentor
2023–2026	Yael Avidan — collaborator/mentee — vegetation data analysis & modeling
2021	Paula Schreiner — B.Sc. thesis, Univ. Tübingen — experimental-design advisor

PROFESSIONAL SERVICE & OUTREACH

Statistical consulting & collaboration (2017–present). Provide study-design, mixed/hierarchical-model, SEM and data-analysis support to 20+ research groups across ecology, conservation and the life sciences. Several collaborations have produced co-authored publications — including with the labs of **Prof. Tamir Klein** (Weizmann Institute), **Prof. Merav Seifan** (BGU) and **Dr. Liora Shaltiel-Harpaz** (MIGAL / Tel-Hai). Paid statistical consulting for the labs of **Prof. Oded Berger-Tal** and **Prof. Carmi Korine** (BGU) and for organizations including the Jewish National Fund (KKL).

Field surveys. Annual monitoring of protected wildflowers — *Iris atropurpurea* (Royal Iris) and *Paeonia* (Forest Peony), Upper Galilee — organized by Dr. Hagai Shemesh (2018, 2021, 2023, 2025, 2026).

AI & computational tools in education. Built interactive simulations of classical ecological models for Prof. Seifan's Community Ecology course; a ChatGPT-based botanical-identification tool for Prof. Giladi's courses (2025); contributing to AI integration in the new international B.Sc. in Renewable Natural Resources, Sde Boker (2026).

Committees & service. Organizing committee, SIDEER symposia, BGU (2018–2020); Academic Committee, Student Council, Sde Boker (2016–2020); session moderator, BGU student conference (2025); curriculum design, new B.Sc. in Bio-Data / Computational Biology, Tel-Hai (2021).

Public outreach. "Introduction to Science" with high-school students (2019–2020); ecology education with elementary-school children (2018).

TECHNICAL & COMPUTATIONAL SKILLS

Statistical modeling. LM/GLM/LMM/GLMM; variance partitioning & hierarchical models and SEM / path analysis (*specialties*); bootstrap & resampling; phylogenetic comparative methods (PGLS, phylogenetic path analysis); multivariate ordination (PCA, NMDS, cluster); spatial & time-series analysis; simulation / individual-based modeling; experimental design.

Programming & reproducibility. R / RStudio (advanced — Shiny, R Markdown / Quarto); Python (data pipelines); NetLogo; Matlab; Git version control. SPSS, JMP/Jamovi, AMOS, Tableau.

GIS & imaging. QGIS, ArcMap; ImageJ (morphological trait quantification).

Field & laboratory. Designed and ran large-scale common-garden (net-house) experiments and multi-year seed-collection field campaigns on the heterocarpic annual *Crepis sancta* (MSc & PhD) — ~1,440 seeds / ~760 plants from 180 field-monitored maternal plants across 4 populations, two irrigation regimes; permanent 50×100 m plots; trait variance partitioned across four nested scales (population → mother → fruit → seed). Also: multi-year arid/Mediterranean vegetation & trait surveys; controlled pollination & wind-dispersal experiments; molecular-biology & analytic-chemistry methods.

Research software development. Designed and built **multiple multi-module R Shiny dashboards** for integrating, exploring and visualizing large multi-source datasets (each with an automated data pipeline and test suite), a Python data-processing pipeline with API integration, and an iOS (Swift) app. Demonstrates end-to-end research-software capability: reproducible workflows, automated pipelines, interactive analytics, and version control across complete projects.

ADDITIONAL TRAINING

2018	Research visit (~1 month), lab of Dr. Pierre-Olivier Cheptou, CEFE Montpellier, France (<i>Crepis sancta</i> population genetics)
2017	Agent-Based Modelling with NetLogo (Univ. of Göttingen); Quantitative Methods in Ecology — advanced multivariate statistics (BGU)
2015	Biostatistics: ANOVA & Design of Experiments (BGU)

LANGUAGES

Hebrew (native) · English (near-native, C2 — academic writing, teaching, and international presentations)

REFERENCES

Available upon request.